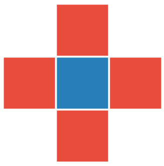
Polyominoes with a Saturated Cell

A saturated cell is a cell all four of whose edge-neighbors are present, so it sits fully surrounded in the interior. `a(n)` is the number of free polyominoes of `n` cells (counted up to rotation and reflection) that contain at least one saturated cell. For $n = 5$ there is a single shape, the plus-pentomino, shown centered. For $n = 6, 7, 8$ EVERY distinct shape is drawn in a grid under one `a(n)` heading, each labelled with its bounding box, so the whole family the count measures is visible – the shapes are genuinely different, not rotations of one another. Ordinary cells are red; the saturated cell is blue.

$$a(5) = 1,\ a(6) = 2,\ a(7) = 9,\ a(8) = 36$$

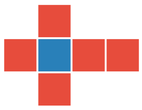
Computed by Peter Exley, Jun 2026

$a(5) = 1$ **[PROVED]** the only shape, 3 x 3 bbox



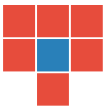
$a(6) = 2$ **[PROVED]** all 2 shapes, bbox under each

3 x 3

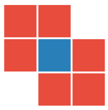


3 x 4

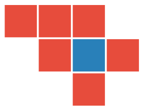
$a(7) = 9$ **[PROVED]** all 9 shapes, bbox under each



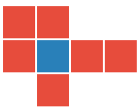
3 x 3



3 x 3

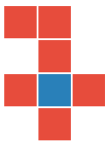


3 x 4

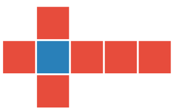


3 x 4

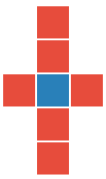
3 x 4



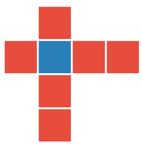
4 x 3



3 x 5



5 x 3



4 x 4

$a(8) = 36$ **[PROVED]** all 36 shapes, bbox under each